5. Golden Hospital Discharge Record Design

You are a **Data Engineer** at a hospital tasked with transforming fragmented clinical data from its **Electronic Health Record (EHR)** system into a single **Gold-Standard Analytical Dataset**. The source data is highly unstructured, inconsistent, and messy.

Your goal is to create a **Golden Discharge Record** table where **each row represents one unique discharge event (IPID)**, and all available structured information is cleanly represented as columns.

The Data Provided

You have been given three raw CSV files containing disconnected operational data:

- **discharge_master.csv** – Admission details, patient IDs, dates, and financial summaries.
- **discharge_details.csv** – Itemized breakdowns of charges, amounts, and procedures.
- **clinical_findings.csv** – Free-text medical notes and treatment descriptions.

Assessment Tasks

1. **Data Preparation and Transformation**
    - Understand and clean each dataset to handle missing values, duplicates, and inconsistent identifiers.
    - Merge the files logically using **IPID** as the central key.
    - Extract key features from clinical notes (e.g., keywords, diagnoses, procedures) to enrich the dataset.
2. **Final Deliverables**
    - A single **analytical table or spreadsheet** where each row represents one **discharge event (IPID)** and all relevant details are cleanly included.
    - A short **document** (max one page) describing the major cleaning and transformation steps you performed.
    - An optional **code script** (SQL, Python, or any tool) if you used automation for cleaning or merging.

Format

- You may use **Excel, SQL, Python (Pandas), or any ETL tool** you're comfortable with.
- Keep your column names meaningful and standardized (e.g., Total_Amount, Diagnosis_Category, Stay_Duration_Days).

Evaluation Criteria

Criteria	Description
Logic and Accuracy	Are the joins, cleaning, and transformations logically sound?
Completeness	Does the final table capture maximum usable information from all files?
Documentation Clarity	Are the cleaning and transformation steps easy to understand?
Data Quality	Are duplicates, missing data, and inconsistencies properly handled?